Smart Parking System for Smart

Prepared For

Smart-Internz

Internet of

Things (IoT)

By

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On

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1. Project Overview

Smart Parking System for Smart Cities is an Internet of Things (IoT) based parking management solution designed to improve parking efficiency in urban environments.
The rapid growth of vehicles in metropolitan areas has increased the demand for intelligent parking systems that can reduce traffic congestion and minimize the time spent searching for available parking spaces.
The proposed system provides a user-friendly web interface that allows users to monitor parking slot availability in real time, reserve parking spaces, and generate QR-based parking tickets. The system utilizes Firebase Realtime Database to synchronize parking information and ensure that slot status is updated instantly across the application.
The project integrates modern web technologies including HTML, CSS, JavaScript, and Firebase Realtime Database to create an interactive dashboard capable of displaying parking status, vehicle statistics, entry and exit gate information, and booking details.
The solution demonstrates how IoT-enabled smart parking systems can contribute towards the development of smart city infrastructure and improved urban mobility.

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2. Key Features

- Real-time parking slot monitoring
- Parking slot reservation system
- QR-based ticket generation
- Vehicle entry tracking
- Vehicle exit tracking
- Parking fee calculation
- Dashboard analytics
- Entry gate monitoring
- Exit gate monitoring
- Responsive user interface
- Firebase cloud integration

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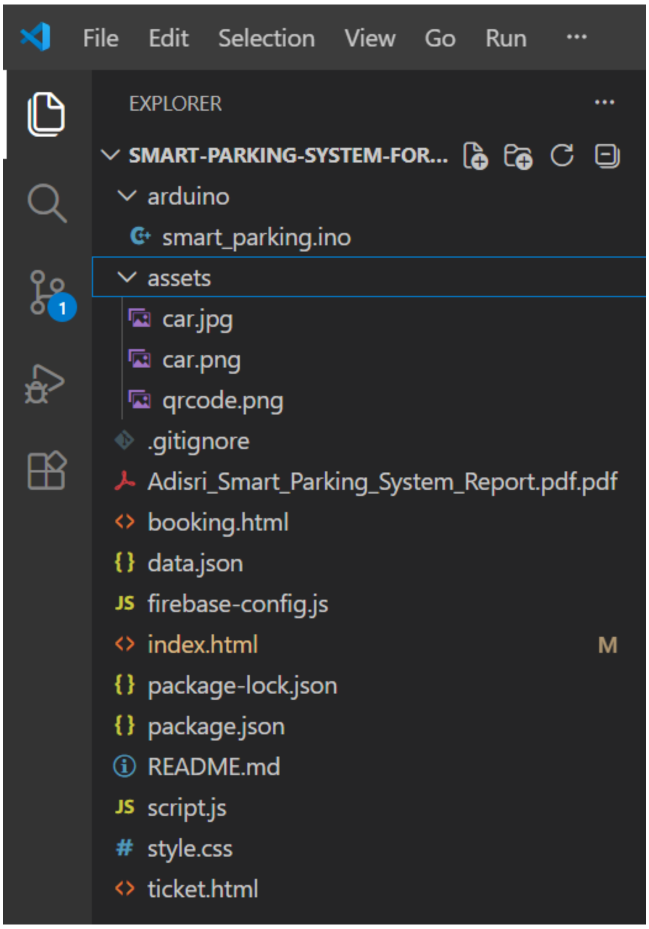
3. Architecture / Folder Structure

```
Smart-Parking-System/
│
├── arduino/
│   └── smart parking.ino
│
├── assets/
│   ├── car.jpg
│   ├── car.png
│   └── qrcode.png
│
├── booking.html
├── data.json
├── firebase-config.js
├── index.html
├── package.json
├── package-lock.json
├── script.js
├── style.css
├── ticket.html
│
└── README.md
```

3.1 Architecture



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4. System Workflow

4.1 Requirement Analysis
- Studied the problem of parking congestion in urban areas.
- Identified the need for a real-time parking management solution.
- Defined project objectives including parking slot monitoring, online booking, and ticket generation.

4.2 User Interface Development
- Designed the Smart Parking Dashboard using HTML, CSS, and JavaScript.
- Developed responsive pages for parking monitoring, booking, and ticket generation.
- Created a modern interface for improved user experience.

4.3 Firebase Realtime Database Integration
- Configured Firebase Realtime Database.
- Established real-time communication between the dashboard and database.
- Stored parking slot information and booking records.

4.4 Parking Slot Monitoring Module
- Implemented parking slot status management.
- Displayed slot availability as FREE, BOOKED, or OCCUPIED.
- Updated parking information dynamically on the dashboard.

4.5 Vehicle Booking Module
- Developed a booking form for vehicle owners.
- Collected user details, vehicle number, start time, and end time.
- Validated booking information before confirmation.

4.6 Parking Fee Calculation

- Implemented automatic parking fee calculation.
- Calculated charges based on booking duration.
- Displayed total parking charges before confirmation.

4.7 QR Ticket Generation

- Generated a unique booking ID for every reservation.
- Created QR-based parking tickets containing booking details.
- Displayed ticket information including vehicle number, slot number, booking time, and payment details.

4.8 Vehicle Entry and Exit Monitoring

- Implemented vehicle count monitoring.
- Displayed total vehicles entered and exited.
- Displayed entry gate and exit gate status on the dashboard.

4.9 GitHub Deployment

- Created a GitHub repository for project management.
- Uploaded project files and documentation.
- Maintained version control throughout development.

4.10 Testing and Validation

- Tested parking slot availability updates.
- Verified booking functionality.
- Validated QR ticket generation.
- Tested dashboard responsiveness on different screen sizes.
- Confirmed successful Firebase synchronization and data updates.

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5. User Scenarios

Scenario 1: Parking Slot Reservation

User: Vehicle Owner

Process:

- The user accesses the Smart Parking Dashboard.
- The system displays the current parking slot status.
- The user selects an available parking slot and proceeds to the booking page.
- The user enters personal details, vehicle number, start time, and end time.
- The system calculates the parking fee automatically.
- The booking is confirmed and stored in Firebase Realtime Database.
- A unique booking ID and QR-based parking ticket are generated.

Scenario 2: Parking Administrator Monitoring Parking Status

User: Parking Administrator

Process:

- The administrator accesses the Smart Parking Dashboard.
- The dashboard retrieves live parking data from Firebase Realtime Database.
- The administrator monitors parking slot status such as FREE, BOOKED, and OCCUPIED.
- Vehicle entry and exit statistics are displayed on the dashboard.
- Entry gate and exit gate status are continuously monitored.
- Any changes in parking slot occupancy are reflected instantly on the dashboard.

Scenario 3: QR Ticket Verification and Parking Access

User: Registered Parking User

Process:

- After completing the booking process, the user receives a QR-based parking ticket.
- The ticket contains booking details including booking ID, vehicle number, parking duration, and parking fee.
- The user presents the QR ticket at the parking facility.
- The booking details are verified using the ticket information.

- The system confirms the reservation and grants access to the allocated parking slot.

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6. Challenges Faced

- Understanding and implementing Firebase Realtime Database.
- Designing a responsive and visually appealing dashboard.
- Developing a reliable booking module.
- Implementing automatic parking fee calculation.
- Integrating QR-based ticket generation.
- Managing parking slot state transitions between FREE, BOOKED, and OCCUPIED.
- Testing multiple booking scenarios.
- Deploying the project to GitHub and maintaining proper documentation.

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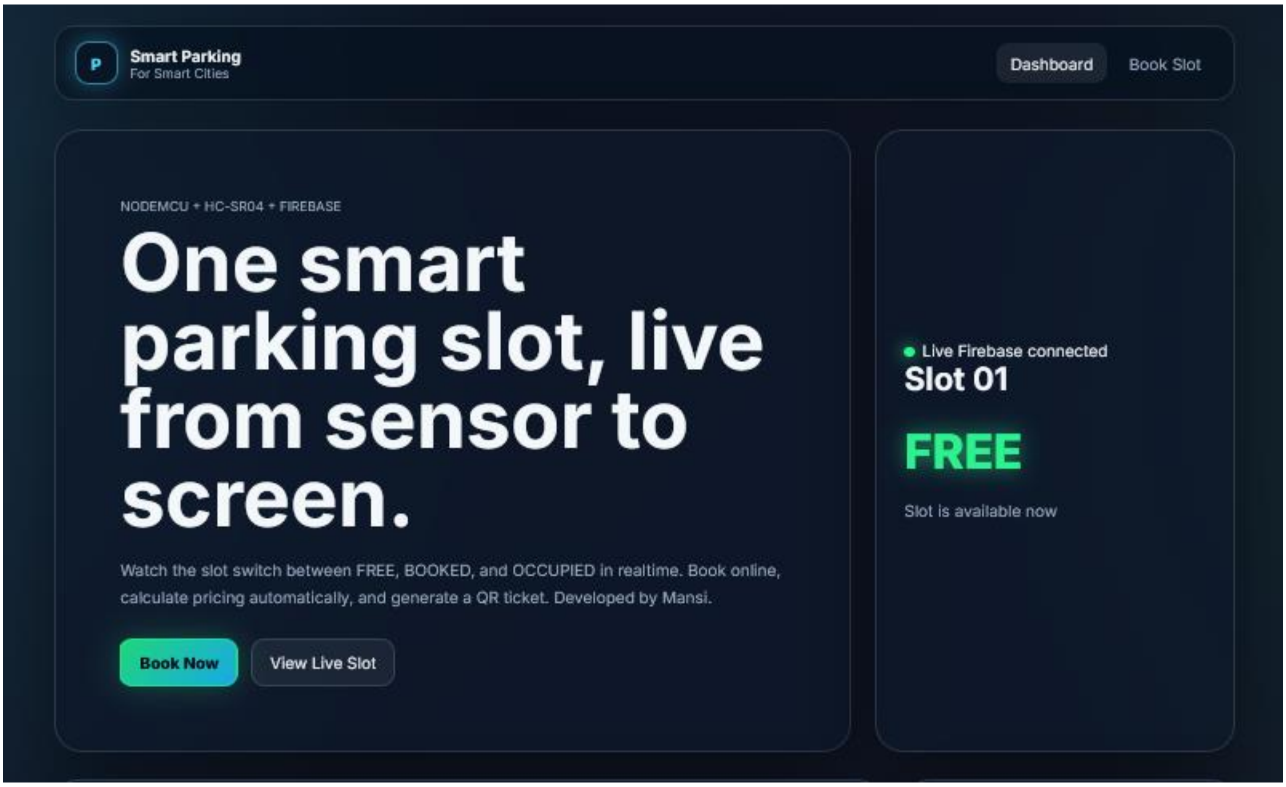
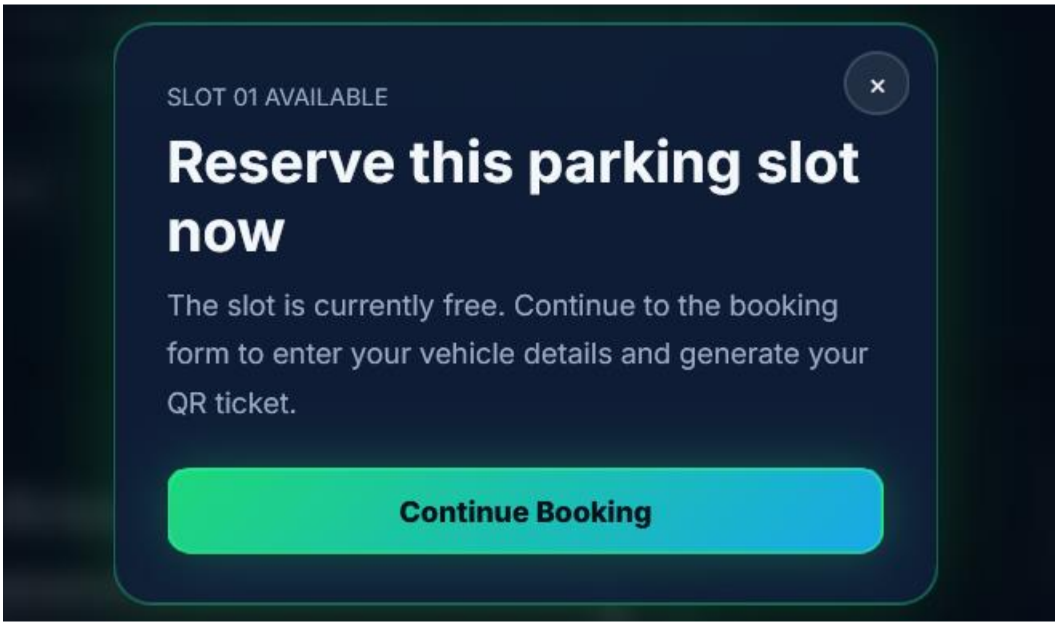
7. Output Pages

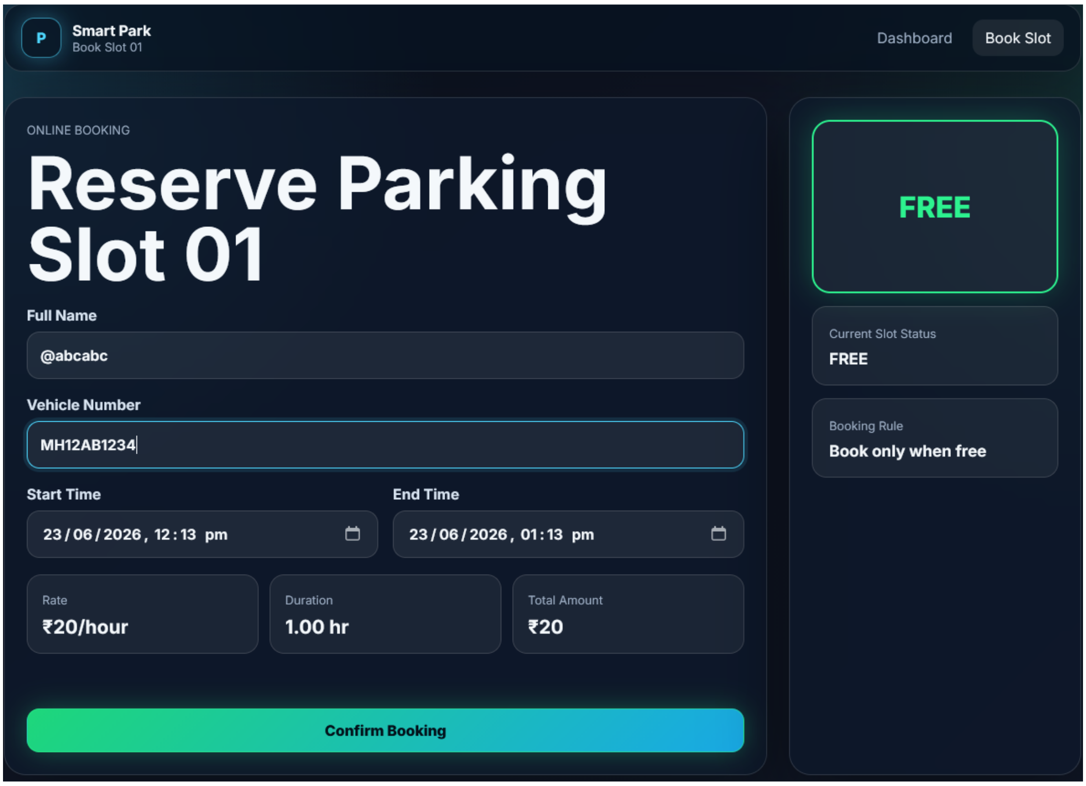
Home Dashboard



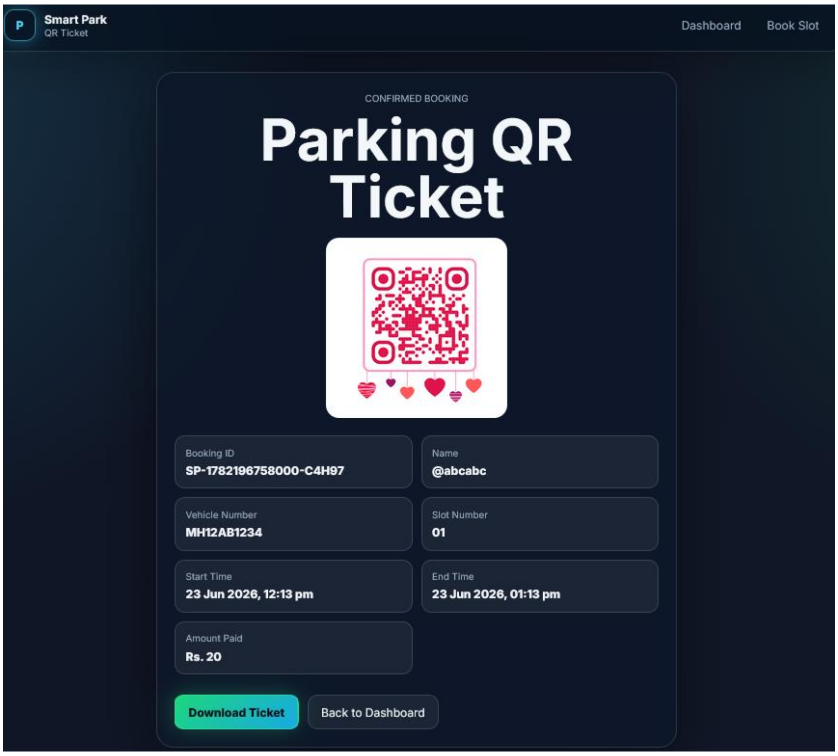
Smart Parking Analytics Panel

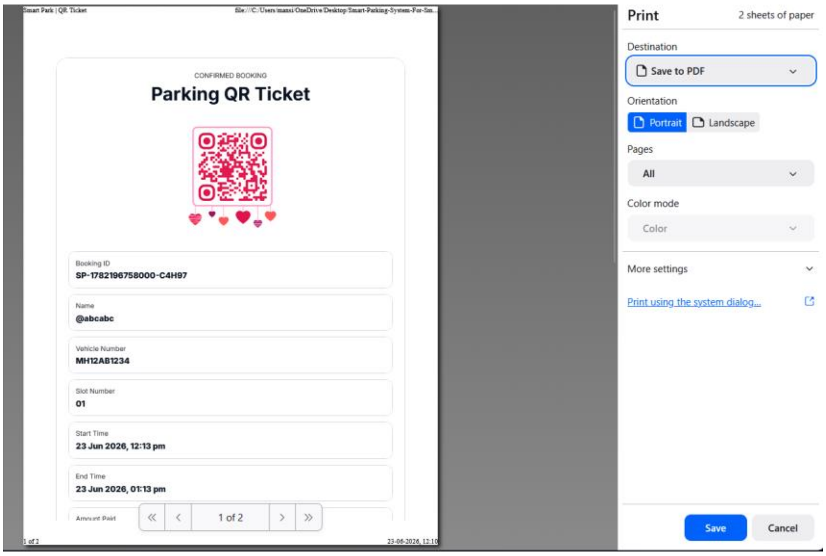


Parking Slot Reservation Space

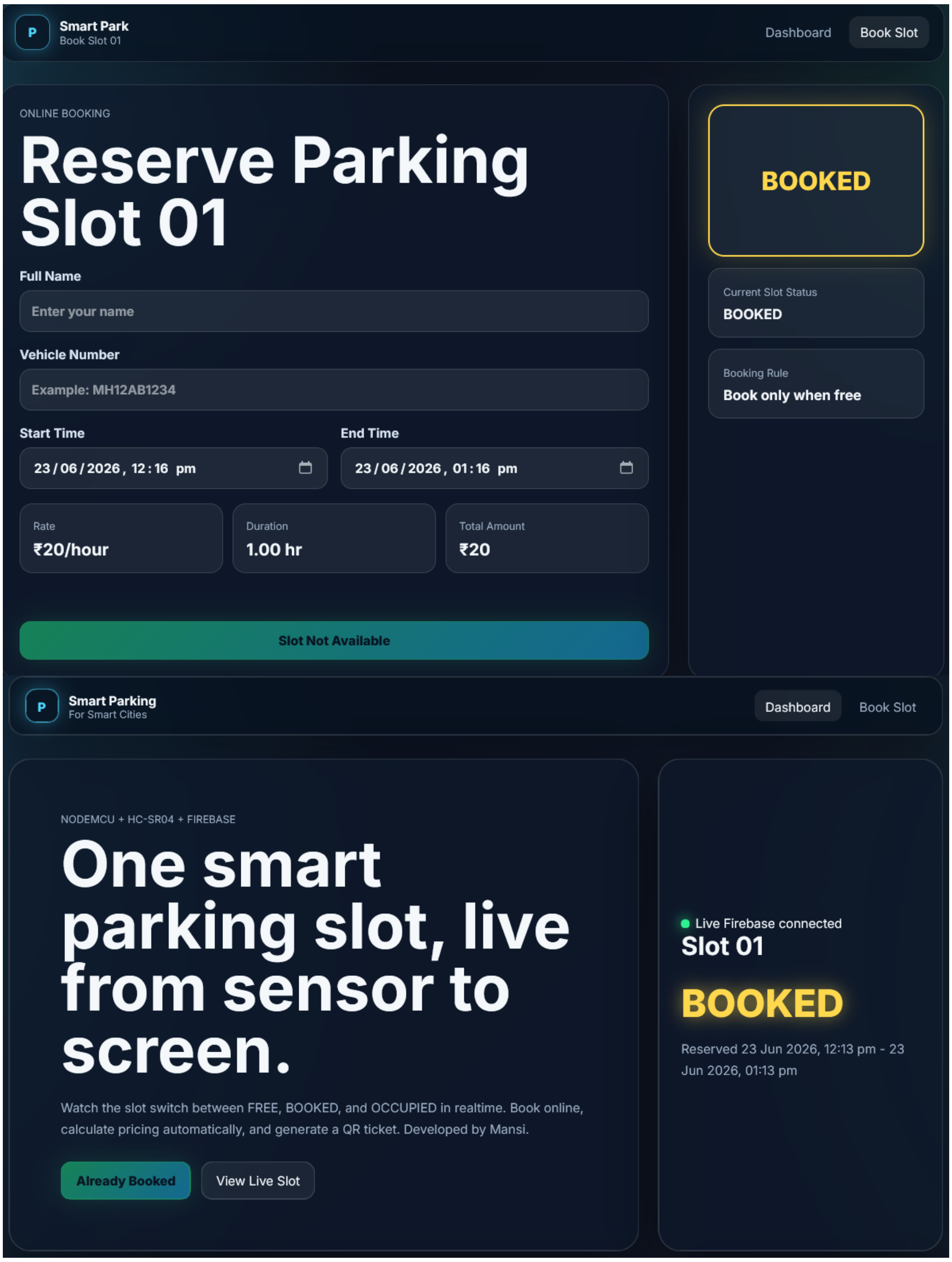


Booking Confirmation QR Code

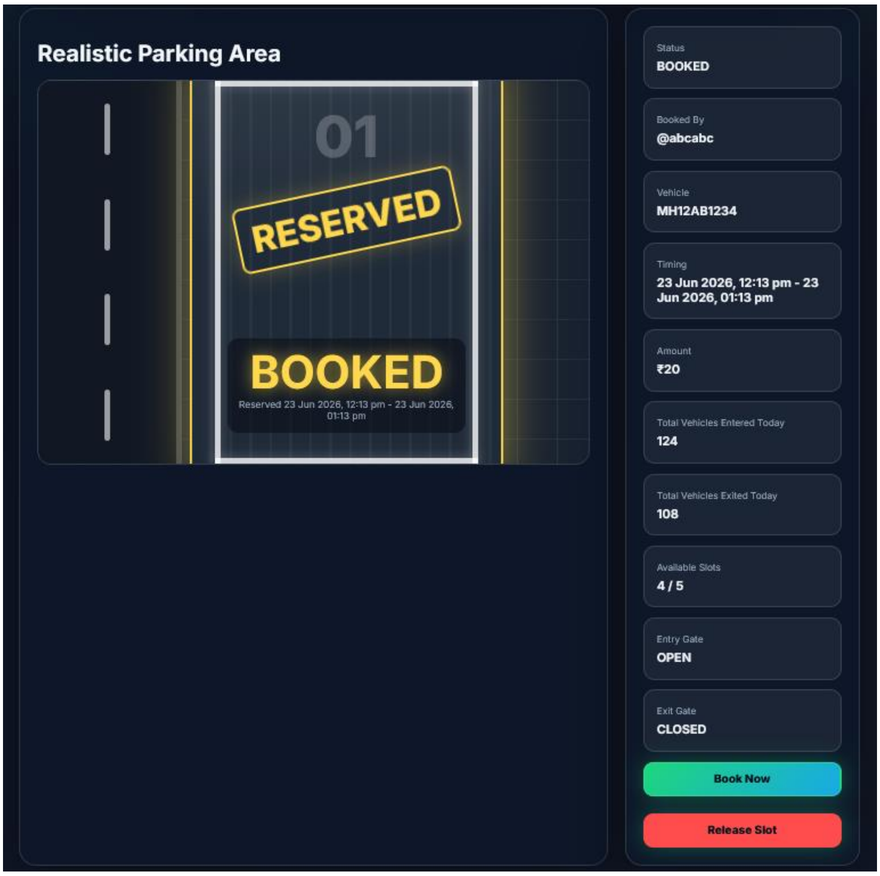




Slot Booked


P
Smart Park
Book Slot 01
Dashboard
Book Slot
ONLINE BOOKING
Reserve Parking Slot 01
Full Name
Enter your name
Vehicle Number
Example: MH12AB1234
Start Time
23/06/2026, 12:16 pm
End Time
23/06/2026, 01:16 pm
Rate
₹20/hour
Duration
1.00 hr
Total Amount
₹20
Slot Not Available
BOOKED
Current Slot Status
BOOKED
Booking Rule
Book only when free
P
Smart Parking
For Smart Cities
Dashboard
Book Slot
NODEMCU + HC-SR04 + FIREBASE
One smart parking slot, live from sensor to screen.
Watch the slot switch between FREE, BOOKED, and OCCUPIED in realtime. Book online, calculate pricing automatically, and generate a QR ticket. Developed by Mansi.
Already Booked
View Live Slot
Live Firebase connected
Slot 01
BOOKED
Reserved 23 Jun 2026, 12:13 pm - 23 Jun 2026, 01:13 pm

View Live Slot



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8. Technologies Used

Layer	**Technology**
Frontend	HTML, CSS, JavaScript
Database	Firebase Realtime Database
IoT Hardware	NodeMCU ESP8266
Sensors	HC-SR04 Ultrasonic Sensor
Version Control	Git & GitHub
Hosting	Firebase Services

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9. Future Enhancements

- Multiple parking zone support
- Automatic vehicle detection using IoT sensors
- License plate recognition
- Mobile application development
- Secure online payment integration
- Cloud analytics dashboard
- Smart parking notifications
- Advanced vehicle tracking system

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10. Conclusion
The Smart Parking System for Smart Cities successfully demonstrates the application of Internet of Things (IoT) concepts, cloud-based databases, and modern web technologies in solving real-world urban parking challenges.
The system provides an efficient platform for monitoring parking slot availability, managing parking reservations, generating QR-based tickets, and tracking vehicle movement through an interactive dashboard.
By integrating HTML, CSS, JavaScript, Firebase Realtime Database, NodeMCU ESP8266, and HC-SR04 Ultrasonic Sensors, the project achieves real-time synchronization of parking information while maintaining a user-friendly experience.
The project highlights how smart parking solutions can reduce traffic congestion, minimize the time spent searching for parking spaces, and improve resource utilization in urban environments. It also demonstrates the potential of IoT-enabled systems in supporting the development of smart cities and intelligent transportation infrastructure.
Overall, the Smart Parking System provides an effective, scalable, and innovative approach to modern parking management and smart city development.

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11. GitHub Repository
GitHub Link:
Mansi04-cloud/Smart-Parking-System-For-Smart-Cities
Demo Video Link:
https://drive.google.com/file/d/1_7OHOpQMc7BjpiqinGR_nRtgg-pLG_4D/view?usp=sharing